

7 QC Tools Training 2 Days

Essential Problem-Solving Tools for Quality Improvement

Duration: 2 Days

Mode: Online / Classroom / On-Site

Certification: QLSS Certificate of Completion



Course Introduction

The 7 QC Tools are the fundamental techniques used for problem-solving, process analysis, and quality improvement. This 1-day training is designed to help professionals across quality, manufacturing, and process teams understand and apply these tools in a structured, data-driven way.

The training blends theoretical understanding with hands-on practice using real-world examples and exercises that can be directly applied on the shopfloor or in quality reviews.



Pre-requisites

- Basic understanding of process workflows or quality control
- Exposure to operational or quality-related tasks is beneficial
- No prior training in statistics or QC tools required



Learning Objectives

- Understand the purpose and application of each QC Tool
- Use tools to identify root causes and analyze defects or failures.
- Generate and interpret charts and graphs to track performance
- Apply tools for decision-making, trend analysis, and standardization
- Solve real-life production and process problems using structured techniques



Course Agenda

Day 1 – Tool Introduction & Basic Application

1. Introduction to Quality and Problem Solving
 - Quality definitions, variation, and role of data
 - PDCA cycle and the role of QC tools in improvement

2. Check Sheet
 - Purpose and types
 - Designing a usable check sheet for data collection
 - Exercise: Create a check sheet for defect tracking

3. Histogram
 - Understanding distribution and variation
 - Creating and interpreting histograms
 - Exercise: Plot histogram using collected data

4. Pareto Chart
 - Prioritizing problems using the 80/20 rule
 - Using Pareto to identify top issues
 - Exercise: Generate Pareto chart using defect frequency

Day 2 – Root Cause, Analysis, and Monitoring Tools

5. Cause and Effect Diagram (Fishbone/Ishikawa)
 - Structure and key categories (5M + 1E)
 - Facilitating brainstorming and RCA
 - Exercise: Build cause-effect diagram for sample issue

6. Scatter Diagram
 - Identifying correlation between two variables
 - Visual representation of potential root causes
 - Exercise: Plot and interpret scatter diagram

7. Control Charts
 - Types: X-bar R, p, np charts
 - Interpreting trends, control limits, special causes
 - Exercise: Build and analyze a control chart

8. Flow Chart (Process Mapping)

- Understanding process flow and decision points
- Identifying inefficiencies and rework
- Exercise: Create flowchart for a real process

Group Activity:

- End-to-end application of 4–5 tools on a sample case
- Group presentations and trainer feedback



What You'll Receive

- QLSS Certificate of Completion
- PDF Training Handout with examples and templates
- 7 QC Tools Worksheet (Printable/Editable)
- Individual performance feedback from trainer



Who Should Attend

- Quality Engineers and Supervisors
- Production and Process Operators
- Maintenance and TPM Teams
- Continuous Improvement and Lean Teams
- Internal Auditors and Quality Circle Members



Trainer Profile

The session is led by experienced quality improvement professionals with deep expertise in Lean Manufacturing, problem-solving, and shopfloor implementation of quality tools. Trainers bring practical insights from solving hundreds of real-life issues using data-driven QC methods.



Instructional Methodology

- Tool-by-tool walkthrough using visual presentations

- Case studies based on actual defects, quality issues
- Hands-on problem-solving using templates and data
- Group discussions and trainer-guided analysis
- Final evaluation of understanding and participation



What Makes QLSS QC Tools Training Unique

- Hands-on practice using real data from manufacturing settings
- Focus on actionable, easy-to-use methods for shopfloor teams
- Delivered by trainers with real experience in root cause analysis and Lean QA
- Customizable to industry-specific quality issues (automotive, pharma, engineering)
- Tools aligned with IATF 16949 and ISO 9001 improvement requirements.